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Qian; Nanjian et al.

Method of resyncing a conserving battery-powered electronic device with an access point, and electronic device and system for performing the method

Abstract

A method of conserving a battery-powered electronic device is disclosed. The electronic device is wirelessly communicable with an access point over a wireless local area network. Also disclosed are an electronic device and a system for performing the method. The method includes causing a first control architecture to be in an idle state; while the first control architecture is in an idle state and with a second control architecture, receiving a beacon frame from the access point, the beacon frame including electronic device association information, checking the beacon frame if the electronic device is associated with the access point, and providing a wakeup signal to the first control architecture when the electronic device is dissociated. The method further includes communicating from the second control architecture to the first control architecture that the electronic device was dissociated from the access point.

Inventors: Qian; Nanjian (San Diego, CA), McGill; Chris (San Diego, CA), Van Hoorebeke; Mark (Carlsbad, CA), Van Cleave; Todd (San Marcos, CA)

Applicant: Arlo Technologies, Inc. (Carlsbad, CA)

Family ID: 88647277

Assignee: Arlo Technologies, Inc. (Carlsbad, CA)

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Primary Examiner: Dagneu; Mekonnen D

Attorney, Agent or Firm: Boyle Fredrickson, S.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application Ser. No. 63/420,947, filed Oct. 31, 2022, the entire contents of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a system and method of conserving battery power in an electronic device wirelessly communicating over a network with an access point. The present invention more particularly relates to a system and method of identifying a non-synced electronic device with an access point and rectifying the same.

2. Discussion of the Related Art

(2) Wi-Fi™ enabled video cameras are in wide use and are often used for monitoring and security purposes. Content captured by wireless enabled cameras can be communicated over the Internet. The content can be viewed in substantially real time and/or recorded for later viewing.

(3) Some video cameras are Wi-Fi™ enabled and battery powered. An example of such a camera is disclosed in U.S. Pat. No. 9,713,084, assigned to Netgear, Inc., the content of which is incorporated herein by reference. A wireless enabled, battery powered camera may have a divided control architecture to help conserve battery life. A first control architecture may have a first functionality that consumes more power and a second control architecture may have a second functionality that consumes less power. Further, the camera may utilize one or more sleep modes to help conserve battery life for the wireless camera. For example, the first control architecture may have a longer sleep mode and can be woken by the second control architecture upon receiving certain communications, and the second control architecture can be woken periodically for maintenance communication.

(4) The wireless camera can be connected to a Wi-Fi™ enable base station (BS). The camera and the base station should be in a “synced” state under normal situations. A synced state refers to both the camera and the base station knowing that the camera is associated with the base station. Some abnormal conditions (e.g., network interference) can cause the camera and the base station to become “out-of-sync” or in an “out-of-sync” state. The camera and the base station may not realize they are out-of-sync for a period of time. When this situation occurs, known past systems are only re-associated when the camera performs a new power cycle synchronization. This causes time delays and increased power losses. Accordingly, a need exists to detect and address an out-of-sync state more quickly.

SUMMARY OF THE INVENTION

(5) According to aspects of the invention, an end device or node uses beacon frames from an access point to more quickly identify an out-of-sync error. A beacon frame is a small packet of data transmitted from the access point (such as a base station) to the nodes. The end device (such as a battery-powered wireless camera) can include a first control architecture (e.g., a camera host hardware chip set) and a second control architecture (e.g., a camera Wi-Fi™ system on chip). The second control architecture periodically acquires the beacon frame while the first control architecture is in an idle state. The access point passes end device association information through the beacon frames. The second control architecture decodes the beacon frame, including whether the association is present. If the association is not present, an out-of-sync error condition arises, and the second control architecture wakes the first control architecture. The end device can then proceed to re-sync with the access point. Accordingly, the end device, such as a Wi-Fi™ enabled camera, can identify and correct an out-of-sync state more quickly than past cameras, thereby improving system performance.

(6) In one aspect, a method of conserving a battery-powered electronic device is disclosed. The electronic device is wirelessly communicable with an access point over a wireless local area network. The method includes causing a first control architecture to be in an idle state; while the

first control architecture is in an idle state and with a second control architecture, receiving a beacon frame from the access point, the beacon frame including electronic device association information, checking the beacon frame if the electronic device is associated with the access point, and providing a wakeup signal to the first control architecture when the electronic device is dissociated. The method further includes communicating from the second control architecture to the first control architecture that the electronic device is dissociated from the access point.

(7) In a further aspect, the method can include, with the first control architecture and in response to being dissociated from the access point, initiating a connection session to the access point and syncing the electronic device with the access point. The method can also include causing the first control architecture to reenter the idle state; and while the first control architecture has reentered the idle state and with the second control architecture, receiving a second beacon frame from the access point, the second beacon frame including the electronic device association information, checking the beacon frame if the electronic device is associated with the access point.

(8) Another aspect of the present invention provides a digital video camera in wireless communication with an access point via a wireless network. The camera includes a lens, an image capture element, an antenna, and a first control architecture. The first control architecture includes a first processor, and a first memory communicably coupled to the first processor. The first memory stores a sleep module including instructions that, when executed by the processor, cause the processor to cause the first control architecture to be in an idle state. The camera further includes a second control architecture. The second control architecture includes a second processor and a second memory communicably coupled to the second processor. The memory stores a communication module including instructions that, when executed by the second processor, cause the second processor to, while the first control architecture is in the idle state, receive a beacon frame from the access point, the beacon frame including electronic device association information, check the beacon frame if the camera is associated with the access point. The second memory further stores an interrupt module including instruction that, when executed by the processor, cause the second processor to initiate a wakeup signal to the first control architecture when the electronic camera is dissociated from the access point.

(9) In a further aspect, the present invention provides a system including a base station coupled to the camera.

(10) These and other features and advantages of the invention will become apparent to those skilled in the art from the following detailed description and the accompanying drawings. It should be understood, however, that the detailed description and specific examples, while indicating preferred embodiments of the present invention, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the present invention without departing from the spirit thereof, and the invention includes all such modifications.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Preferred exemplary embodiments of the invention are illustrated in the accompanying drawings in which like reference numerals represent like parts throughout, and in which:

(2) FIG. 1 is block diagram representing a system for capturing, transmitting, and displaying images according to aspects of the invention;

(3) FIG. 2 is a block diagram representing a camera shown in FIG. 1;

(4) FIG. 3 is a block diagram representing a base station shown in FIG. 1; and

(5) FIG. 4 is an example flow chart of a method of operating the system of FIG. 1.

(6) FIG. 5 is an example timing diagram for communication among the base station and camera of FIG. 1.

(7) FIG. 6 is an example beacon frame transmitted from the base station to the camera in the system of FIG. 1.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(8) FIG. 1 represents a system **10** for capturing, transmitting, and displaying images, such as images of a video. The video is acquired at a first location and displayed at a second location. The shown system **10** includes a wireless camera network **15** having a plurality of wireless node cameras **20A-C** transmitting and receiving wireless signals to and from an access point (AP) via the wireless network **25**. The access point is shown as a base station **30**. Although three cameras **20A-C** are shown in FIG. 1, a system can have more or fewer cameras. The system **10** also includes a remote electronic device, which is shown as a mobile user device **35**, and a video stream service **40**. The user device **35** and the video stream service **40** are in communication with the base station **30** via a gateway **45** and the Internet **30**. It is contemplated that many different arrangements are possible for the system **10**, and the shown system **10** is provided for ease of explanation. For example, the base station **30** may be one component of a larger wired and/or wireless local area network (LAN) having many access points communicating with the cameras **20A-C**.

(9) The cameras **20A-C** are in communication with the base station **30**. The base station **30** can host a private local area network, which is shown as the wireless local area network (WLAN) **55**. The WLAN **55** of one implementation is an IEEE 802.11 local area network (LAN) that follows the network protocols of the Wi-Fi™ Alliance. Such a network is typically referred to as a Wi-Fi™ network. However, the WLAN **55** can be or include other wireless local area networks.

(10) Electronic devices communicate data to and from the cameras **20A-C**. One example electronic device is the mobile user device **35**. Exemplary mobile user devices can be a smart phone, tablet computer, or a laptop computer, etc. The electronic device can alternatively be a stationary user device. A stationary user device, such as a desktop computer, is an electronic device that is generally considered by one skilled in the art as stationary even though the device can move.

(11) Another electronic device that communicates data to and from the camera is the video stream service **40**. The video stream service **40** can be a commercially available service for storing and/or analyzing the images and/or video. Exemplary analyzing services include enabling the camera to differentiate between humans and animals, reducing false motion notifications, sending alerts and screenshots to the user device **35**, etc. The video stream service **40** can also provide a secure gateway for video to be communicated from the cameras **20A-C** to the user device **35**. An exemplary video stream service **40** is Arlo's Smart™ video stream service available from Arlo Technologies, Inc. in San Jose, California, U.S.

(12) FIG. 2 represents an example camera **20** shown in FIG. 1. In the illustration, the camera **20** has a small and compact housing **70** for enclosing and protecting the various camera components illustrated as blocks in FIG. 2. The camera **20** includes a lens **75** and an image capture element (or primary sensor) **80**. The image capture element **80** can be any suitable type of image capturing device or sensor, including, for example, an area array sensor, a Charge Coupled Device (CCD) sensor, a Complementary Metal Oxide Semiconductor (CMOS) sensor, or a linear array sensor, just to name a few possibilities. The image capture element **80** may capture images in suitable wavelengths on the electromagnetic spectrum. The image capture element **80** may capture color images and/or grayscale images.

(13) The camera **20** has a field of view **85** extending radially from the outwardly facing lens **75**. The field of view **80** is a portion of the environment **90** within which the camera **20** can detect electromagnetic radiation via the lens **75** and image capture element **80**. The camera **20** is configured to capture images. An image is a digital representation of a scene from the environment **90** as captured by the camera **20**. Capturing an image refers to the act of obtaining and recording an image data file or stream of the digital representation. The scene is the portion of the environment **90** observed through the field of view **85**. Capturing a plurality of images in a timed sequence can result in a video. Capturing a video refers to the act of obtaining and recording a video data file or

stream of the digital representation.

(14) Referring back to FIG. 2, the camera **20** has a controller **95** including a first control architecture **100** and a second control architecture **105**. The first control architecture **100** includes a first processor **110** and a first memory **115**, and the second control architecture **105** includes a second processor **120** and a second memory **125**. While the arrangement of FIG. 2 shows the controller having dual processors and memories, it is envisioned that many other arrangements are possible.

(15) The processors **110** and **120** can include any component or group of components that are configured to execute, implement, and/or perform any of the processes or functions described herein or any form of instructions to carry out such processes or cause such processes to be performed. Examples of suitable processors include a microprocessor, microcontroller, and other circuitry that can execute software. Further examples of suitable processors include, but are not limited to, a central processing unit (CPU), an array processor, a vector processor, a digital signal processor (DSP), a field-programmable gate array (FPGA), a programmable logic array (PLA), an application specific integrated circuit (ASIC), programmable logic circuitry, and a controller. The processors **110** and/or **120** can include a hardware circuit (e.g., an integrated circuit) configured to carry out instructions contained in program code.

(16) The memories **115** and **120** store one or more types of instructions and/or data. The memories **115** and **120** can include volatile and/or non-volatile memory. Examples of suitable memory include RAM (Random Access Memory), flash memory, ROM (Read Only Memory), PROM (Programmable Read-Only Memory), EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), registers, disks, drives, or any other suitable storage medium, or any combination thereof. The memories **115** and/or **120** can be a component of a processor, can be operatively connected to a processor for use thereby, or a combination of both.

(17) The memories **115** and **125** can include various instructions stored thereon. For example, the memories **115** and **125** can store one or more modules. Modules can be or include computer-readable instructions that, when executed by a processor, cause a processor to perform the various functions disclosed herein. Example modules in memory **115** include a sleep module for causing the first control architecture to enter and exit an idle state. Example modules in memory **125** include a communication module for communicating with the base station **30**, and an interrupt module for initiating a wakeup signal to the first control architecture. While functions may be described herein for purposes of brevity, it is noted that the functions are performed by the first processor **110** or the second processor **120** using the instructions stored on or included in the various modules described herein. Some modules may be stored remotely and accessible by a processor using, for instance, various communication devices and protocols.

(18) In accordance with one construction, the first control architecture **100** includes a separate device from the second control architecture **105** such that the first and second control architectures can enter and exit idle states independently of one another. As used herein, an idle state is broadly defined as stand by, suspend, power off, hibernation, hybrid sleep, sleep, and related modes. In one construction, the first control architecture **100** includes camera host hardware and the second control architecture **105** includes communication system-on-chip (SoC) hardware. The first control architecture **100** (e.g., the camera host hardware) performs image processing functions on image data received from the image capture element **80**, and controls handling of messages received from and transmitted to the base station **30**. When there are no messages to be received or transmitted, and no image processing functions to be performed, the first control architecture **100** can be powered down or placed in a low-power sleep mode. The second control architecture **105** (e.g., the communication system-on-chip hardware) controls operation of the camera **20** when the first control architecture **100** is powered down and powers up the first control architecture **100** upon the occurrence of one or more predetermined triggering events. The first control architecture **100**,

when awake, is the master controller of the camera **20** and functions as the host processor to the radio (discussed below) and the image capture element **80**. Further discussion regarding the relationship between the first control architecture and the second control architecture will be provided below.

(19) Before moving to other components of the camera **20**, it should be understood by somebody skilled in the art that the controller **95** includes many additional conventional elements typically found in a wireless camera controller. Example elements include, but are not limited to, one or more floodlights or any of various sensors and detectors employed by surveillance cameras. Further discussion regarding these components is not provided herein since the components are conventional.

(20) The camera **20** communicates wirelessly (e.g., with the base station **30**) via a radio **127**. An example of a radio includes a wireless local area network (WLAN) radio. With the WLAN radio **127**, the camera **20** generally communicates over a short-range wireless communication network, such as the WLAN **55**. In one implementation, the radio **127** includes a transceiver **130** for transmitting and receiving signals to and from the base station **30**, via an antenna **135**. The transceiver **130** can be separate to or part of the second control architecture **130**. The wireless communication can be as prescribed by the IEEE 802.11 standards in accordance with the Wi-Fi™ communication protocol. It is appreciated, however, that the camera **20** can be adapted to perform communications in accordance with any known or to be developed communication protocol, or even a proprietary communication protocol developed for a particular application.

(21) The camera **20** is battery-powered by a battery (or battery pack) **140**. In one implementation, the life of the battery **140** is extended by having the camera **20** normally operate in an idle state, and only activating the camera, or portions of the camera, for necessary periods of time to perform one or more desired functions. For example, the default mode of operation of the camera **20** is a sleep mode wherein the controller **95** is substantially deactivated. In this mode, the controller **95**, or any deactivated portion thereof, are only activated on an “as needed” basis. Further examples of this operation will be described below.

(22) The camera **20** can further include secondary sensors **145**, which may also be used to generate an interrupt signal. For example, a secondary sensor **145** may be a microphone which generates an interrupt signal upon receiving an audio signal above a defined threshold. Alternatively, the secondary sensor **145** may be a motion sensor for sensing motion in a room, for example. Such secondary sensors **145** are useful, for example, in security situations to detect if an unauthorized entry into a particular area has occurred. Upon detecting a substantial noise and/or motion, an interrupt signal is generated, which can then activate a portion of the controller **95**. Other example secondary sensors **145** include a temperature sensor, an image sensor, and a vibration sensor.

(23) An exemplary camera capable of incorporating aspects of the invention is an Arlo Ultra brand camera available from Arlo Technologies in San Jose, California, US. Before moving to other components of the system **10**, it should be understood by somebody skilled in the art that the camera **20** includes many additional conventional components typically found in a wireless camera. Example components include, but are not limited to, floodlights, doorbells, and any of various sensors. Further discussion regarding these components is not provided herein since the components are conventional.

(24) Turning now to FIG. **3**, the figure represents an example of the base station **30** shown in FIG. **1**. In the illustration, the base station **30** has a housing **150** for enclosing and protecting the various components illustrated as blocks in FIG. **3**. The base station **30** has a controller **152**, including a processor **155** and a memory **160**. While the arrangement of FIG. **3** shows a single processor **155** and a single memory **160**, it is envisioned that many other arrangements are possible. For example, multiple elements of the base station **30** can include a distinct processor and memory.

(25) The processor **155** can include a component or group of components that are configured to execute, implement, and/or perform any of the processes or functions described herein for the base

station **30** or a form of instructions to carry out such processes or cause such processes to be performed. Examples of suitable processors include a microprocessor, a microcontroller, and other circuitry that can execute software. Further examples of suitable processors include, but are not limited to, a core processor, a central processing unit (CPU), a graphical processing unit (GPU), an array processor, a vector processor, a digital signal processor (DSP), a field-programmable gate array (FPGA), a programmable logic array (PLA), an application specific integrated circuit (ASIC), math co-processors, and programmable logic circuitry. The processor **155** can include a hardware circuit (e.g., an integrated circuit) configured to carry out instructions contained in program code.

In arrangements in which there are a plurality of processors, such processors can work independently from each other or one or more processors can work in combination with each other.

(26) The base station **30** includes a memory **160** for storing one or more types of instructions and/or data. The memory **160** can include volatile and/or non-volatile memory. Examples of suitable memory include RAM (Random Access Memory), flash memory, ROM (Read Only Memory), PROM (Programmable Read-Only Memory), EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), registers, disks, drives, or any other suitable storage medium, or any combination thereof. The memory **160** can be a component of the processor **155**, can be operatively connected to the processor **155** for use thereby, or a combination of both.

(27) In one or more arrangements, the memory **160** can include various instructions stored thereon. For example, the memory **160** can store one or more modules. Modules can be or include computer-readable instructions that, when executed by the processor **155**, cause the processor **155** to perform the various functions disclosed for the module. While functions may be described herein for purposes of brevity, it is noted that the functions are performed by the processor **155** using the instructions stored on or included in the various modules. Some modules may be stored remotely and accessible by the processor **155** using, for instance, various communication devices and protocols.

(28) The base station **30** also could be combined with a gateway router **45** or another device in a single module, which would still be considered a “base station” within the meaning of the present disclosure. Stated another way, the access point formed by the base station could be contained in the gateway router **45** and/or other system component(s).

(29) The base station **30** communicates wirelessly (e.g., with the cameras **20A-C**) via a radio **165**. An example of a radio includes a wireless local area network (WLAN) radio. With the WLAN radio, the base station **30** generally communicates over a short-range wireless communication network, such as the WLAN **55**. In one implementation, the radio **165** includes a transceiver **170** for transmitting and receiving signals to and from the camera **20**, via an antenna **175**. The transceiver **170** can be separate to or part of the controller **152**. The wireless communication can be as prescribed by the IEEE 802.11 standards in accordance with the Wi-Fi™ communication protocol. It is appreciated, however, that the base station **30** can be adapted to perform communications in accordance with any known or to be developed communication protocol, or even a proprietary communication protocol developed for a particular application.

(30) The base station **30** includes the user interface **180**. The user interface **180** can include an input apparatus and an output apparatus. The input apparatus includes a device, component, system, element, or arrangement or groups thereof that enable information/data to be entered into the base station **30** from a user. The output apparatus includes any device, component, or arrangement or groups thereof that enable information/data to be presented to the user. The input apparatus and the output apparatus can be combined as a single apparatus, such as a touch screen commonly used by many electronic devices.

(31) The base station **30** includes a communication port **185**, which is configured to provide a communication interface between a larger computer network, such as the Internet **50** via the gateway **45**, for example, and the wireless camera network **15**.

(32) In one construction, since the base station **30** is powered by an enduring power source (e.g., power outlet), it is not necessary for the base station **30** to be operated in a default sleep mode, although this is not precluded. An exemplary base station capable of incorporating aspects of the invention is an Arlo SmartHub brand base station available from Arlo Technologies in San Jose, California, US. Before moving to the operation of the system **10**, it should be well understood by somebody skilled in the art that the base station **30** includes many additional conventional components typically found in a base station or access point.

(33) For the camera **20** described above, it should be appreciated that conserving the power of the battery **140** and avoiding undue power drainage of the battery **140** is highly desirable. By conserving power and avoiding undue power drainage, the battery **140** requires less frequent recharging and/or replacement, which is not only more convenient but also less expensive for the user.

(34) During operation of the system **10**, the base station **30** broadcasts a beacon frame, among other communication. A beacon frame is one of the management frames in IEEE 802.11 based WLANs. A beacon frame contains information about the network and is transmitted periodically. The beacon frame serves to announce the presence of a wireless LAN and to synchronize the members of the service set. Beacon frames are transmitted by the base station **30** in an infrastructure basic service set (BSS).

(35) While beacons do cause some non-trivial overhead for a network, they are necessary for the proper operation of a network. When a wireless node (e.g., camera **20**) receives a beacon frame, it receives information about the capabilities and configuration of that network. The wireless node may also be able to generate a list of available eligible networks, sorted by signal strength.

(36) After initial power ON, the camera **20** attempts to associate with one or more AP's. For example, the camera **20** can associate with the base station **30**. Once associated with the base station **30**, the camera **20** may transition to an idle state. In this idle state, the camera **20** is in a low power state where all non-essential components or units are powered down or in sleep mode in order to conserve power. In one embodiment, the first control architecture **100**, and more specifically the first processor **110**, are placed in an idle state, to be awakened by the second control architecture, and more specifically the second processor **120**, upon the occurrence of a predetermined triggering event. In the idle state, information pertaining to the last state of the controller **95** is maintained in the first memory **115**, as necessary, so that when the first control architecture is awakened, it may resume operations from its last state before it entered the idle state.

(37) After associating with a network, such as network **55**, the second control architecture **105** of the camera **20** continues to scan for beacons. This has several benefits. For example, as the camera **20** still receives beacon frames from the currently associated base station **30**, the camera **20** is able to use the timestamps in those beacon frames to update its internal clock. Beacon frames from the currently associated base station **30** also inform the camera **20** of imminent configuration changes.

(38) Another benefit of beacons is to enable the camera **20** to have power saving modes. For example, the base station **30** holds on to packets destined for the nodes that are currently idled. In a traffic indication map of a beacon frame, the base station **30** is able to inform nodes, such as camera **20**, that they have frames waiting for delivery.

(39) In one operation, the camera **20**, when in the idle state, periodically listens for beacons from the base station. The camera **20** may skip a predetermined number of beacons then attempt to capture the next beacon. Contained in the beacon is a Traffic Indication Map (TIM) or a Delivery Traffic Indication Map (DTIM), which contains information indicating whether information or a message is waiting to be sent to the camera **20**. If the TIM or DTIM indicates that the camera **20** has a message waiting to be sent to the camera **20**, this will trigger the second processor **120** to wake the first processor **110**, which will then request the message or data from the base station **30**. Upon receiving the message, the first processor **110** will process any instructions or data contained in the message and, thereafter, the camera **20** returns to the idle state with the first control

architecture **100** and other non-essential units powered down or in a sleep mode. In one construction, the first processor **110**, the first memory **115**, and image capture element **80** are devices that are powered down when the camera **20** is associated with the base station **30** but in an idle state. In this state, the second processor **120**, the second memory **125**, and the radio **127** are placed in low-power modes.

(40) If an event occurs that requires the attention of the first processor **110** and the first processor **110** is already powered up, then the second processor **115** will simply inform the first processor of an interrupt. If the first processor **110** is in an idle state when the event occurs, the second processor **120** first awakes the first processor and then informs it of the interrupt. When the first processor **110** has completed all of its operations it will again power down and enter an idle state. Before powering down, the first processor **110** saves its current state information, and any other desired information, in the first memory **115**.

(41) Accordingly in some operations, the second processor **120** only wakes the first processor **110** when the camera needs to transmit traffic (e.g., images) to the base station **30**, or when the second processor **120** receives layer three traffic packets (such as an IP packet) from the base station **30**. When the camera **20** is connected to the base station **30**, the camera **20** and the base station **30** should be in the synced state in a normal situation (meaning both sides know that the camera is connected). Some abnormal conditions can cause the camera **20** and the base station **30** to go out of sync and not realize the components are out of sync for a very long time. For example, the camera **20** may believe it is in a connected state but the base station **30** may believe the camera **20** is not associated anymore.

(42) One implementation for detecting and addressing an out-of-sync situation is shown in FIG. 4. At the base station **30** side, the processor **155** maintains an association list in the memory **160** (block **200**). The association list refers to nodes that are associated or connected to the base station. The base station **30** periodically communicates the current association list in the form of association information. For example, the base station **30** can periodically broadcast the camera MAC address list associated with the base station **30** (block **210**). As another example, the base station **30** and cameras **20A-C** can be from the same manufacture/vendor. Under this scenario, the base station **30** can use shorthand proprietary lists, or even bits, to communicate association information. The base station **30** creates, defines, and encodes a specific information element (IE) to include the camera association information (block **205**). The base station **30** will include this AP specific IE into the beacon frame.

(43) At the camera **20**, the second control architecture **105** will wake up from its idle state (block **215**), receive the beacon frame (block **220**) decode the AP specific IE, and retrieve the node association information (block **225**). The processor **120** determines whether the camera **20** is identified in the association information (block **230**). If the processor **120** determines that the camera **20** is dissociated from the base station **30**, the processor **120** clears its state machine locally, and wakes the first control architecture **100** (block **235**) to inform it of the camera dissociation condition (block **240**).

(44) The first control architecture **100** upon receiving the disassociation condition cleans its state machine. The processor **110** will then start a new connection session (block **245**) to associate the camera **20** to the base station **30** and re-sync with the base station **30** again.

(45) FIG. 5 provides a diagram showing the flow of packets and actions between the base station **30** and the camera **20**. In this example, the camera **20** has been associated to the base station **30** for a time and the first control architecture **100** is asleep. The camera **20** has been assigned a bit (e.g., the first bit) inside the AP specific Information Element (IE). The IE is embedded in the payload of the beacon frame that the base station **30** broadcasts. As illustrated, the camera **20** receives a continuous stream of beacon messages from the base station **30**. In the first two illustrated beacon messages, the first bit inside the AP specific IE is on. This is indicated by the hexadecimal value 0x01. If the bit assigned for a particular camera is true, then the corresponding camera **20** receiving

the beacon message knows that the base station **30** has marked it as associated. When the camera **20** detects that it is identified as associated by the base station **30**, nor further action is required by the camera. In the third illustrated beacon message, the first bit inside the AP specific IE is off. This is indicated by the hexadecimal value 0x00. If the bit assigned for a particular camera is false, then the corresponding camera **20** receiving the beacon message knows that the base station **30** has marked it as disassociated. As further illustrated in FIG. 5, when the camera **20** detects that it is identified as disassociated by the base station **30**, the camera **20** takes additional action to rectify this situation. The second control architecture **105** will execute steps **235-245**, as previously discussed, to wakeup up the first control architecture **100** and have the camera **20** establish a new connection session with the base station **30**.

(46) FIG. 6 provides a snap shot of a decoded beacon frame that displays the association IE of the beacon payload. The AP specific IE can be placed anywhere in the AP specific IE list (order does not matter, it could be the first IE or last IE, as long as the base station **30** and the camera **20** know the arrangement). In one embodiment, the IE's contents would be a simple bit mask where each bit in the bit mask is assigned to a specific camera **20A-C** when they first associate.

(47) Accordingly, the invention provides a new and useful method for associating a battery-powered node wirelessly communicable with an access point, detecting a loss of association between the node and the access point, and re-associating the node and the access point. Aspects of certain embodiments described herein may be implemented as software modules or components. As used herein, a software module or component may include any type of computer instruction or computer-executable code located within or on a computer-readable storage medium, such as a non-transitory computer-readable medium. A software module may, for instance, comprise one or more physical or logical blocks of computer instructions, which may be organized as a routine, program, object, component, data structure, etc. that perform one or more tasks or implement particular data types, algorithms, and/or methods.

(48) A particular software module may comprise a single instruction or many instructions, and may be distributed over several different code segments, among different programs, and across several computer-readable storage media. In addition, data being tied or rendered together in a database record may be resident in the same computer-readable storage medium, or across several computer-readable storage media, and may be linked together in fields of a record in a database across a network.

(49) The block diagrams and flowcharts in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments. In some embodiments, one element may be designed as multiple elements or multiple elements may be designed as one element. In some embodiments, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale. It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved.

(50) The systems, components, and/or processes described above can be realized in hardware or a combination of hardware and software and can be realized in a centralized fashion in one processing system or in a distributed fashion where different elements are spread across several interconnected processing systems. A typical combination of hardware and software can be a processing system with computer-usable program code that, when being loaded and executed, controls the processing system such that it carries out the methods described herein. The systems, components, and/or processes also can be embedded in a computer-readable storage, such as a computer program product or other data programs storage device, readable by a machine, tangibly embodying a program of instructions executable by the machine to perform methods and processes

described herein. These elements also can be embedded in an application product which comprises all the maintenance conditions enabling the implementation of the methods described herein and which, when loaded in a processing system, is able to carry out these methods.

(51) Nothing in this application is considered critical or essential to the present invention unless explicitly indicated as being “critical” or “essential.” In addition, the individual components need not be assembled in the disclosed configuration but could be assembled in virtually any configuration. Furthermore, all the disclosed features of each disclosed embodiment can be combined with, or substituted for, the disclosed features of every other disclosed embodiment except where such features are mutually exclusive.

Claims

1. A method of operating an electronic device, the electronic device wirelessly communicable with an access point over a wireless local area network, the method comprising: causing a first control architecture to be in an idle state; while the first control architecture is in an idle state and with a second control architecture: receiving a beacon frame from the access point, the beacon frame including electronic device association information, checking the beacon frame if the electronic device is associated with the access point, and providing a wakeup signal to the first control architecture when the electronic device is dissociated; and communicating from the second control architecture to the first control architecture that the electronic device is dissociated from the access point.
2. The method of claim 1, the method further comprising, with the first control architecture and in response to being dissociated from the access point: initiating a connection session to the access point; and syncing the electronic device with the access point.
3. The method of claim 2, the method further comprising: causing the first control architecture to reenter the idle state; while the first control architecture has reentered the idle state and with the second control architecture, receiving a second beacon frame from the access point, the second beacon frame including second electronic device association information; checking the beacon frame if the electronic device is associated with the access point.
4. The method of claim 3, the method further comprising not providing the wakeup signal to the first control architecture when the electronic device is identified in the beacon frame.
5. The method of claim 1, wherein the electronic device includes a camera, and the access point includes a base station in wireless communication with the camera.
6. The method of claim 1, wherein the beacon frame follows the IEEE 802.11 standard.
7. The method of claim 1, wherein checking the beacon frame includes scanning the electronic device association information for an identifier related to the electronic device.
8. A digital video camera in wireless communication with an access point via a wireless network, the camera comprising: a lens; an image capture element; an antenna; a first control architecture including a first processor, a first memory communicably coupled to the first processor and storing a sleep module including instructions that, when executed by the first processor, cause the first control architecture to be in an idle state; a second control architecture, the second control architecture including a second processor and a second memory communicably coupled to the second processor and storing: a communication module including instructions that, when executed by the second processor, cause the second processor to: while the first control architecture is in the idle state, receive a beacon frame from the access point, the beacon frame including electronic device association information, and check the beacon frame if the camera is associated with the access point, an interrupt module including instructions that are executed by the second processor when the electronic camera is dissociated from the access point, wherein the interrupt module causes the second processor to initiate a wakeup signal to the first control architecture.
9. The digital video camera of claim 8, wherein the first control architecture includes a camera host

hardware, and wherein the second control architecture includes communication system-on-chip (SoC) hardware.

10. The digital video camera of claim 9, wherein the memory of the camera host hardware includes instructions for camera related applications and instructions for a host communication driver, and wherein the SoC hardware includes instructions for wireless communication.

11. The digital video camera of claim 9, wherein the SoC hardware includes instructions for communicating following the IEEE 802.11 standard.

12. The digital video camera of claim 8, wherein the first memory further stores a second communication module having instructions that, when executed by the first processor, cause the first processor to initiate a connection session with the access point in response to being dissociated from the access point, and sync the camera with the access point.

13. The digital video camera of claim 8, wherein the beacon frame follows the IEEE 802.11 standard.

14. A system comprising: a base station comprising: a first antenna, a first processor, and a first memory communicably coupled to the first processor and storing a communication module including instructions that, when executed by the processor, cause the processor to periodically communicate beacon frames, each beacon frame including an information element identifying associated nodes; and a camera comprising: a lens, an image capture element, a second antenna, a first control architecture including a second processor, a second memory communicably coupled to the second processor and storing a sleep module including instructions that, when executed by the second processor, cause the second processor to cause the first control architecture to be in a sleep mode; and a second control architecture, the second control architecture including a third processor and a third memory communicably coupled to the third processor and storing: a communication module including instructions that, when executed by the third processor, cause the third processor to, while the first control architecture is in a sleep mode: receive each beacon frame from the base station, the beacon frame including the information element, and check the information element of the beacon frame to determine if the camera is associated with the access point, and an interrupt module including instruction that, when executed by the third processor, cause the third processor to initiate a wakeup signal to the first control architecture when the electronic camera is dissociated from the access point.

15. The system of claim 14, wherein the beacon frame follows the IEEE 802.11 standard.
